

Compare data sets



1

- a 3 b 4 c 4 d 6
e Group A f Group B

2

- a A b A c B d A
e 35

3

a

	Mean	Range
Class 1	5.2	3
Class 2	8.9	3

- b Class 2, their results are mostly 9 and 10 out of 10, and the mean is 8.9. Whereas, the results for Class 1 are mostly 4 or 5 out of 10. And the mean is only 5.2. which is close to 50%, the result expected if students had simply guessed in a true/false quizz.

4 Humbleton, because the mean is greater than the median.

5

- a Paris b Rome
c New York d Tokyo and Rome
e 0% f 20%

6

- a \$6000 b \$5000 c \$2000 d \$34 000
e \$6000

7

- a Vertical axis b Vertical axis c Athens d Stockholm
e
i False ii True

Box plots

8



9

- a Red bricks have a greater median, so are stronger on average.
- b Yellow bricks have a smaller range, so the strength is more consistent.

10

a

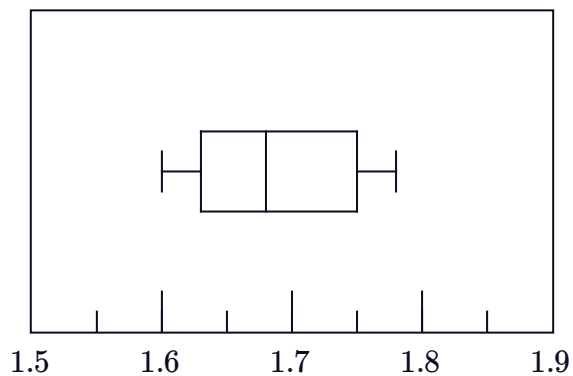
Minimum	1.60
First quartile	1.63
Median	1.68
Third quartile	1.75
Maximum	1.78

b

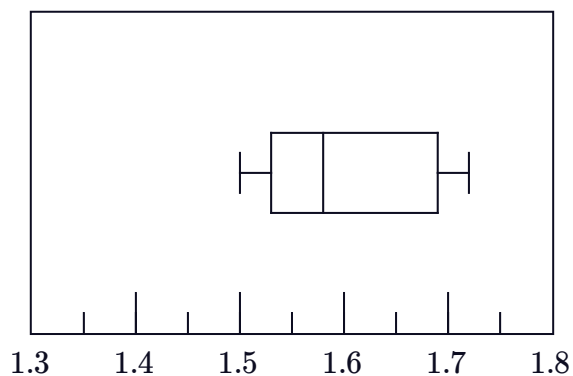
Minimum	1.50
First quartile	1.53
Median	1.58
Third quartile	1.69
Maximum	1.72

c

Height of the boys



Height of the girls



11



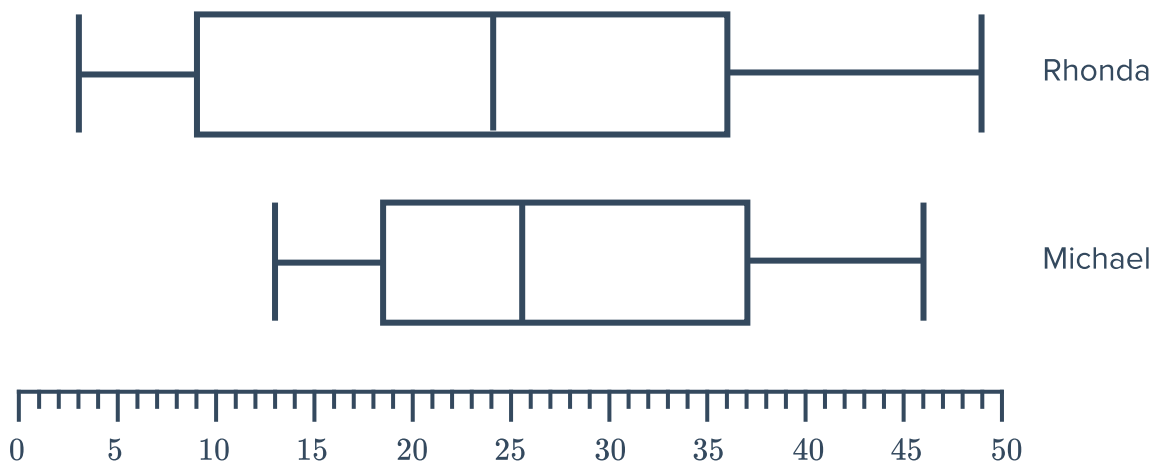
a

Minimum	3
First quartile	9
Median	24
Third quartile	36
Maximum	49

b

Minimum	13
First quartile	18.5
Median	25.5
Third quartile	37
Maximum	46

c



12

a

	Manufacturer A	Manufacturer B
Median	4000	5000
Lower Quartile	2500	3500
Upper Quartile	4500	6000
Range	4000	6500
Interquartile Range	2000	2500

b Manufacturer B

13



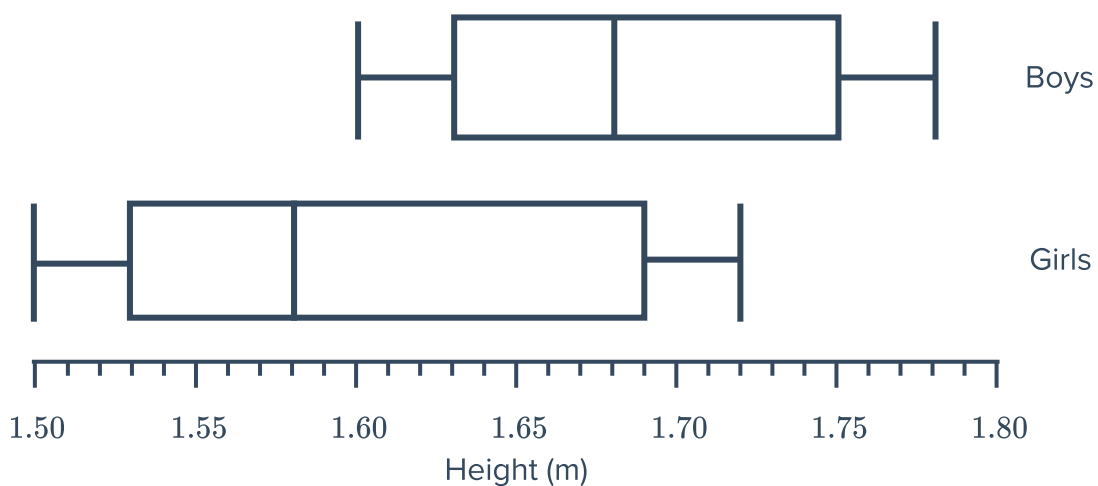
a

Minimum	1.60
First quartile	1.63
Median	1.68
Third quartile	1.75
Maximum	1.78

b

Minimum	1.50
First quartile	1.53
Median	1.58
Third quartile	1.69
Maximum	1.72

c



14

a

	English	Mathematics
Median	10	10
Lower Quartile	8	7
Upper Quartile	12	15
Range	8	15
Interquartile Range	4	8

b Mathematics

15



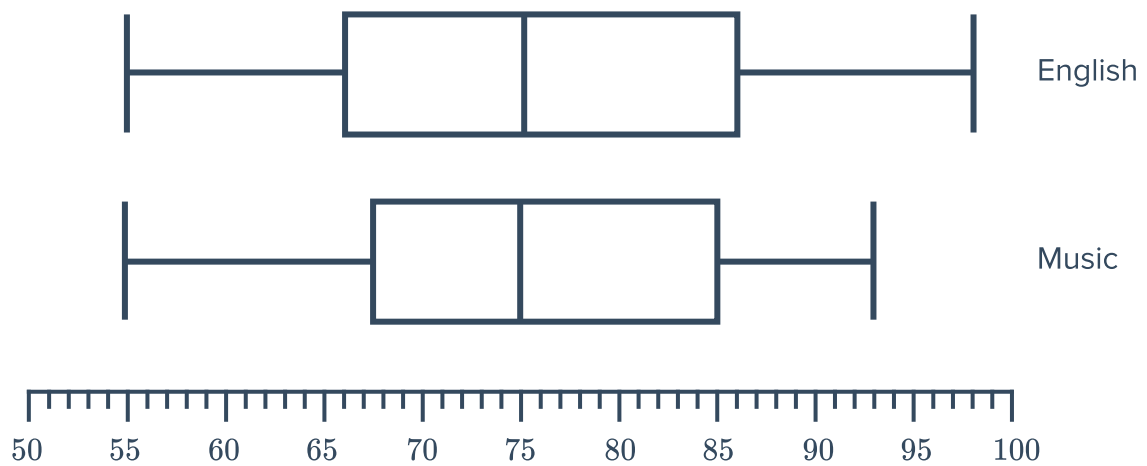
a

Minimum	55
First quartile	66
Median	75.5
Third quartile	86
Maximum	98

b

Minimum	55
First quartile	67.5
Median	75
Third quartile	85
Maximum	93

c



16

a 49

b 50

c 12

d Carl

e Carl

17

a 12 bpm

b 12 bpm

c 12 bpm

d 52 bpm

e 56 bpm

f Before

g 16 bpm

18

a Bookstore A

b \$10

c True

d True

19

a 2010



b 2006

20

a Matt

b Matt

c Matt

d Paul

21

a

	Class 9P	Class 9Q
Median	13	13
Lower quartile	10	11
Upper quartile	18	14
Range	13	11
Interquartile range	8	3

b Class 9P. The top 25% of their scores are between 18 and 20, while the top 25% for 9Q is between 14 and 20.

22

a Sophie

b 1

c 1

d 2

23

a False

b True

c True

d True

e True

24

a Drama

b German

25

a Upmarket restaurant

b \$22

c Upmarket restaurant

d \$36

e 26

26

a 46

b 41

c 23

d 35

e 7

f 26

27



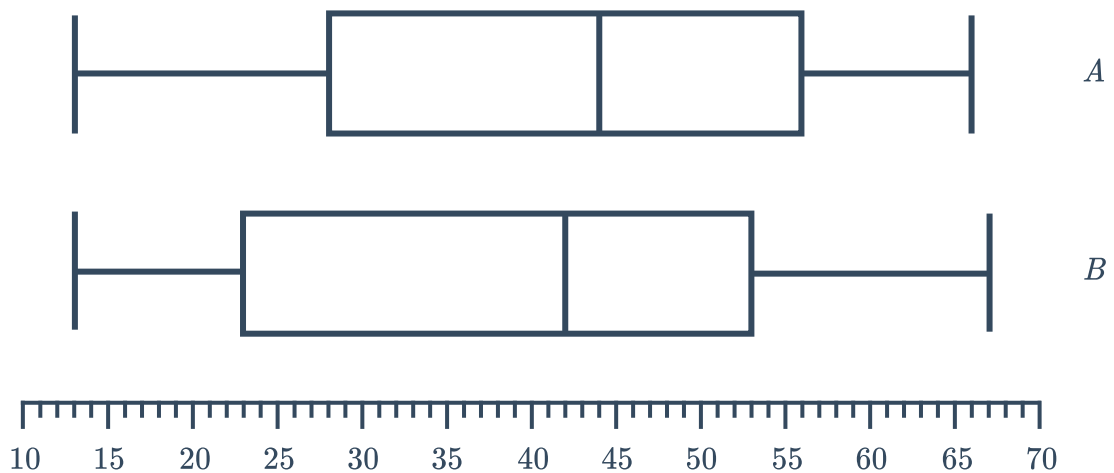
a

Minimum	13
First quartile	28
Median	44
Third quartile	56
Maximum	66

b

Minimum	13
First quartile	23
Median	42
Third quartile	53
Maximum	67

c



28

a Weightlifter A

b The range

c The median

d Weightlifter B

29

a Film C, because all the ages are 18 or older.

b Film B, because the median age is 16.

c Film B, it appeals to a the widest audience.



Back-to-back stem and leaf plots

30

Music		French
9 1	5	9
9 1	6	2 4 6
9 4 1	7	1 3 6
6 6 4	8	2 3 7 9
4 3	9	9

31

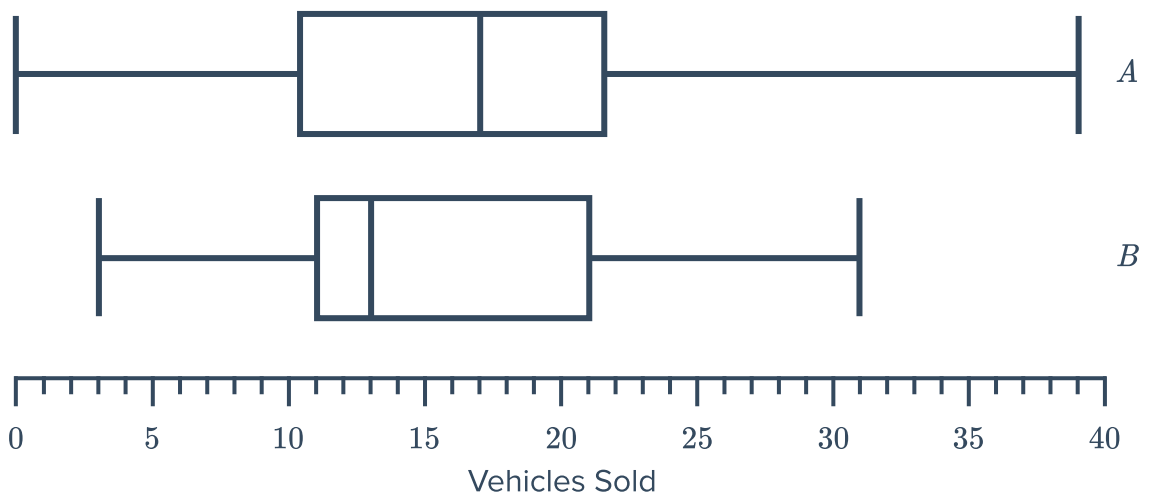
a

Minimum	0
First quartile	10.5
Median	17
Third quartile	21.5
Maximum	39

b

Minimum	3
First quartile	11
Median	13
Third quartile	21
Maximum	31

c



32



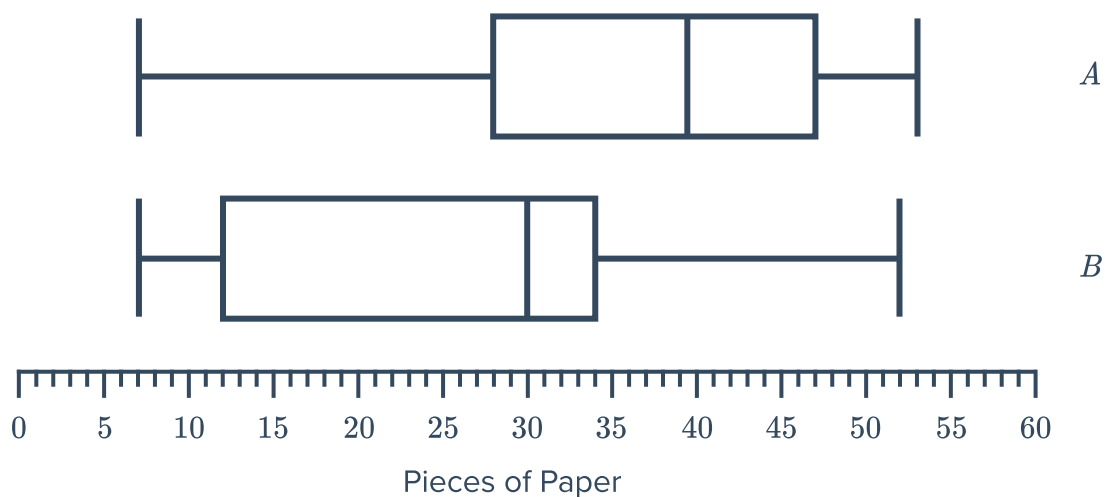
a

Minimum	7
First quartile	28
Median	39.5
Third quartile	47
Maximum	53

b

Minimum	7
First quartile	12
Median	30
Third quartile	34
Maximum	52

c



33

a \$108

b \$89.00

c 65%

d 50%

e 25%



a

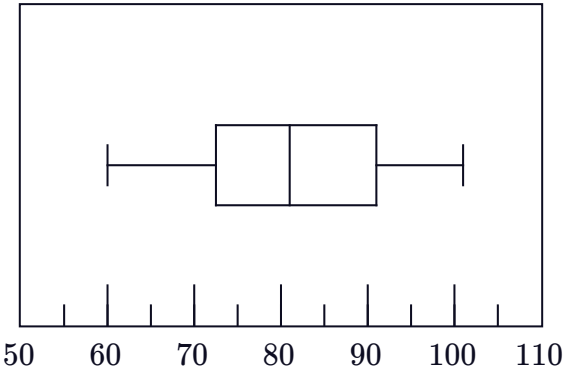
Minimum	\$60
First quartile	\$72.5
Median	\$81
Third quartile	\$91
Maximum	\$101

b

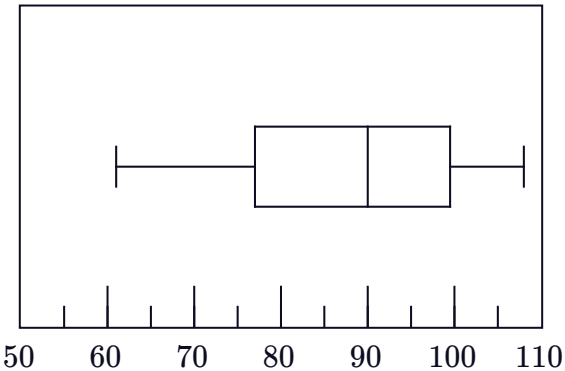
Minimum	\$61
First quartile	\$77
Median	\$90
Third quartile	\$99.5
Maximum	\$108

c

Local prices



International prices



35



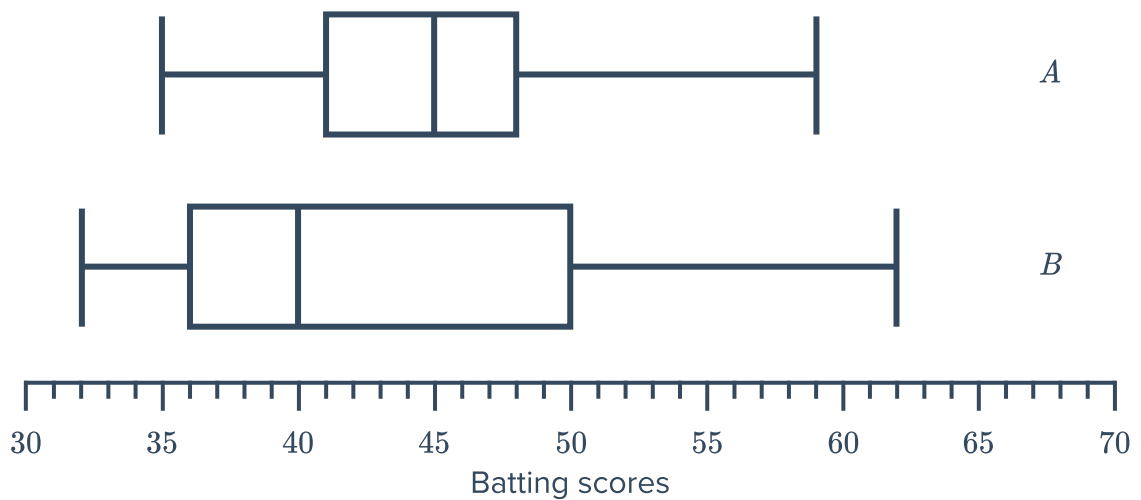
a

Minimum	35
First quartile	41
Median	45
Third quartile	48
Maximum	59

b

Minimum	32
First quartile	36
Median	40
Third quartile	50
Maximum	62

c



36

a

Minimum	50
First quartile	55
Median	68
Third quartile	73
Maximum	80

b

Minimum	84
First quartile	97
Median	108
Third quartile	115
Maximum	121

37

a 7 books

b 24 books

c 30 books

38 a Tricia b 58 mins  c 36 mins d 33 mins

39 a 69.8 kg b 58.7 kg c Men

40 a Tourists: 30 hours, Locals: 33.5 hours b 31.5 hours
c 48 hours

41 a 84 b 94 c 73.1 d 79.0
e 76.32

42 a Hotel A b Hotel B

Tables

43 a 6.5 beans b 6.7 beans
c 7 beans d 4 beans
e Plant B as the mean is greater. f Plant B as the range is smaller.

44 a 17 mm b 16 mm c 29.5 mm d 37.5 mm
e B

45 a 6 b 16 c 3.2 d 3.2

46 a Positively skewed
b Berger's Burgers
c Fry's Fries, because the range is smaller.
d Fry's Fries, because it has a greater median

47

a 86

b 84.8



c 85.4

d Student A

e 72

48

a

	Person 1	Person 2
Mean	7.6	7.4
Median	7	7.5
Mode	7	7
Range	5	1

b Person A because they have the largest range.

c Person A because they have the greatest mean.

49

a

	Men	Women
Mean	77 142.86	72 857.14
Median	80 000	70 000
Mode	80 000	70 000
Range	10 000	10 000

b Men are earning the greater salary as mean, mode and median salaries are all greater for men.

50

a

	Class 1	Class 2
Mean	2.6	8.1
Median	3	8
Mode	3	8
Range	4	4

b Class 2 because it has the greater mean score.